

$c \rightarrow D^0$ fragmentation functions

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The code has three FF: BCFY, KniehlKramer and HymnD.

1 Variables

The fragmentation fraction z_h is the fraction of the charm quark momentum carried by the D^0 ,

$$p_{D^0} = z_h p_c, \quad (1)$$

where p_{D^0} is the transverse momentum we measure (`pD0`) and p_c the one of the charm quark (`pc`). Every integrand is multiplied by

$$\frac{D(z_h, m_T)}{z_h^2}, \quad z_h \in [z_{h,\min}, z_{h,\max}] = [0.05, 1], \quad (2)$$

and z_h is one of the variables of the Monte Carlo integration (VEGAS). The limits are `z_h_min` and `z_h_max` in `src/main.cpp` and `src/main_xpom.cpp`.

The FF is taken at the fragmentation scale

$$m_T = \kappa \sqrt{p_{D^0}^2 + m_c^2}, \quad \kappa = \text{SCALE_FACTOR (default 1)}, \quad (3)$$

with $m_c = 1.5$ GeV. The scale cannot be below m_c : if it is, the code uses $m_T = m_c$. The scale variation uses $\kappa = 0.5$ and 2 .

2 How the FF gets into the calculation

The three FFs are tables in the LHAPDF format (`lhagrid1`), on a grid in z_h and m_T . They are read by the same function, `MakeLHAPDFGridInterpolator(file, flavor, m_T)` in `src/lhapdf_grid.cpp`:

1. it takes the column of the charm quark (PDG id 4),
2. it interpolates linearly in m_T between the two closest grid values. If m_T is outside the grid, it uses the closest grid value,
3. the table has $z_h D(z_h, m_T)$, so it divides by z_h ,
4. it gives back an `Interpolator` for $D(z_h)$ at this m_T .

This is done once per run in `src/main.cpp` (and `src/main_xpom.cpp`), because m_T depends only on p_{D^0} . The result is stored in `param.D_frag_interp`. In `src/integrands.cpp` all the integrands end with the same line:

```
return kernel * fragmentation(z_h, par); // D(z_h) / z_h^2
```

Then, the FF can be changed without modifying the integrands.

3 The fragmentation functions in this code

3.1 BCFY (Braaten–Cheung–Fleming–Yuan)

File `input/BCFY_EK0/bcfy_eko.0000.dat`. At the lowest scale, $m_T = 1.5$ GeV, it is the BCFY function, and it is evolved to higher scales with DGLAP (EKO). The grid has 200 points in z_h from 0.05 to 1 and 30 points in m_T from 1.5 to 50 GeV.

At this scale it is the sum of the part for the D^0 produced directly and the part for $D^{*0} \rightarrow D^0$, with the formulae of Braaten, Cheung, Fleming and Yuan, Phys. Rev. D **51** (1995) 4819 (arXiv:hep-ph/9409316). The parameter is $r = 0.1$ and the weights of the two parts are 0.168 and 0.39.

3.2 Kniehl–Kramer

File `input/KK_EK0/kk_eko.0000.dat`, with the same grid and the same evolution as BCFY. At $m_T = 1.5$ GeV it is the function of Kniehl and Kramer, Phys. Rev. D **74** (2006) 037502 (arXiv:hep-ph/0607306), with $N = 0.694$ and $\epsilon = 0.101$.

3.3 HymnD

File `input/HymnD/prompt-D0-1-109_0000.dat`, the central member of the set of Epele, Hekhorn, Helenius, Paukkunen and Zurita, “Towards new D meson fragmentation functions” (arXiv:2609.10327). The set has 100 more members (replicas), which give the uncertainty band of the FF: `run_scripts/run_members.sh` with `MEMBER.SET=HymnD` runs the calculation once per member. Only the central member is in the repository, the others are in `input/HymnD/` as `prompt-D0-1-109_NNNN.dat`.

4 How to choose the FF

The FF is the fourth argument of the programs,

```
./build/bin/D0 <dipole_file> <pD0> <y> <BCFY|KnihlKramer|HymnD> <channel>
```

and in the workflows it is chosen with environment variables (the defaults are in `run_scripts/config.sh`):

Variable	Default	
FRAG_TYPE	HymnD	FF of one workflow script
FRAGS	all three	FFs of <code>run_all.sh</code>
SCALE_FACTOR	1.0	κ of the fragmentation scale
SCALE_FACTORS	0.5 2.0	values of κ in <code>run_scale_variation.sh</code>
HYMND_FILE	central member	file of HymnD
BCFY_EKO_FILE	input/BCFY_EKO/bcfy_eko_0000.dat	file of BCFY
KK_EKO_FILE	input/KK_EKO/kk_eko_0000.dat	file of KnihlKramer

For example:

```
CHANNEL=0n0n FRAG_TYPE=BCFY ./run_scripts/run_nucleus.sh
```

The results go to a folder with the name of the FF, `output/<CHANNEL>/central_values/<FRAG>/`.

5 How to add another FF

5.1 A table in the LHAPDF format

Example:

```
HYMND_FILE=path/to/my_ff_0000.dat FRAG_TYPE=HymnD ./run_scripts/run_nucleus.sh
```

To add it with its own name:

1. Add the name to `enum class FragmentationType` in `src/def.hpp`.
2. In `src/main.cpp` and `src/main_xpom.cpp`, add the name where `frag_tag` is read, and a line where the FF is built:

```
d_frag_interp = MakeLHAPDFGridInterpolator("input/MyFF/my_ff_0000.dat",
                                           4, frag_scale);
```

3. Add the name to `FRAGS` in `run_scripts/run_all.sh`.

If you want it in the plot of `plotting/scripts/D0_bins.py`, add it to `FRAG_SCHEMES` there too.

The file must have this form (the lines before the first `---` are not read):

```
---
z_h values (one line)
m_T values in GeV (one line)
flavors (one line, it must contain 4)
z_h*D for every flavor, one line per (z_h, m_T): first all m_T for the first z_h, ...
---
```

5.2 A formula

If the FF is a formula and does not depend on the scale, no file is needed. Write a function like `MakeKniehlKramerInterpolator()` that fills two vectors and gives back the interpolator:

```
std::vector<double> z_h_vals(n), dvals(n);
for (int i = 0; i < n; i++) {
    z_h_vals[i] = 0.05 + (1.0 - 0.05) * i / (n - 1.0);
    dvals[i]     = my_D(z_h_vals[i]);
}
return std::unique_ptr<Interpolator>(new Interpolator(z_h_vals, dvals));
```

and then follow the three steps above.

Careful

- The interpolator must give $D(z_h)$, not $z_h D(z_h)$ and not $D(z_h)/z_h^2$. Tables in the LHAPDF format have $z_h D$: `MakeLHAPDFGridInterpolator()` already divides by z_h . The factor $1/z_h^2$ of Eq. (2) is added in `fragmentation()`.
- The grid must cover the range of the z_h integral, from 0.05 to 1.
- The grid in m_T should start at $m_c = 1.5$ GeV or below and go up to $2\sqrt{p_{D^0}^2 + m_c^2}$ for the largest p_{D^0} (about 24 GeV for $p_{D^0} = 12$ GeV). Outside the grid the code does not stop: it uses the closest value of m_T .